

Data validation in Excel is a powerful feature used to control what a user can enter into a cell. It ensures data integrity by preventing errors, maintaining consistency, and guiding the user during data entry.

1. Core Purpose of Data Validation

The primary goal is to restrict the type of data or the values that users enter into a cell. This is essential for:

- **Preventing Errors:** Ensuring that numerical fields don't accidentally contain text.
 - **Standardizing Lists:** Making sure users pick from a predefined set of options (e.g., specific City names) rather than typing them manually, which often leads to typos.
 - **Data Consistency:** Identifying and removing duplicates to keep the dataset clean.
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2. Common Validation Criteria

When setting up data validation, you can choose from several rules to apply to a cell:

- **Whole Number / Decimal:** Restricts entry to numbers within a specific range (e.g., between 1 and 100).
 - **List:** Creates a dropdown menu in the cell. This is the most common use of data validation.
 - **Date / Time:** Ensures the entry is a valid date or time within a specific timeframe.
 - **Text Length:** Limits the number of characters that can be typed into a cell (e.g., exactly 10 digits for a phone number).
 - **Custom:** Allows you to write your own formula to validate the data.
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3. Enhancing User Experience

Data validation isn't just about stopping bad data; it's also about helping the user enter the right data.

- **Input Message:** You can set a message that appears when the user clicks on the cell, explaining what should be entered.
 - **Error Alert:** If a user enters invalid data, you can choose how Excel responds:
 - **Stop:** Prevents the entry entirely.
 - **Warning:** Warns the user but allows them to proceed if they choose.
 - **Information:** Simply informs the user that the data doesn't match the rule.
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4. Advanced Techniques

Excel offers more sophisticated ways to handle and organize validated data:

- **Conditional Formatting for Cleanup:** You can use conditional formatting to highlight existing data that violates validation rules or to find and remove duplicates.
- **Format as Table:** When you format a range as a table, data validation rules are automatically applied to any new rows you add.
- **Subtotals and Summaries:** Validated lists make it much easier to create accurate subtotals and summaries because you don't have to worry about variations in spelling or formatting.

Summary of Steps to Apply Validation

1. Select the cell or range you want to restrict.
2. Go to the **Data** tab and select **Data Validation**.
3. On the **Settings** tab, choose your criteria (e.g., List).
4. Optional: Set an **Input Message** and an **Error Alert**.
5. Click **OK**.

To illustrate how **Data Validation** works in a real-world scenario, here is a detailed example of creating a **Dropdown List** for an employee database.

Practical Example: Employee City Selection

Imagine you are managing an employee sheet like the one seen in **image_20618f.png**. Instead of letting users type the city manually—which could lead to variations like "Bangalore," "Bengaluru," or typos—you want to restrict entries to a specific list.

1. Set Up the Source

First, create a list of valid cities in a separate column or sheet:

- Hubli
- Mysore
- Bangalore
- Mangalore

2. Apply the Validation Rule

1. Select the **City** column cells (e.g., Column E).
2. Navigate to the **Data** tab and select **Data Validation**.
3. In the **Settings** tab, under "Allow," select **List**.
4. In the "Source" box, highlight the list of cities you created in step 1.

3. Configure the User Experience

- **Input Message:** Set the title to "City Selection" and the message to "Please select a city from the dropdown menu". This informs the user immediately upon clicking the cell.
- **Error Alert:** Set the style to **Stop**. If a user tries to type "Delhi," Excel will block the entry and show your custom message: "Invalid City. Please choose from the approved list".

Other Common Examples

Scenario	Validation Criteria	Benefit
Salary Entry	Whole Number (e.g., between 10,000 and 100,000)	Prevents accidental entry of negative numbers or unrealistic values.
Joining Date	Date (e.g., greater than 01-01-2020)	Ensures the date is valid and falls within the company's operational period.
Employee ID	Text Length (e.g., exactly 5 characters)	Standardizes ID formats across the entire organization.

How this helps with "Sorting and Filtering"

Using validation to standardize data makes your **sorting and filtering** much more effective. For instance:

- If you filter for "Hubli", you won't miss any records due to a typo like "Hubly".
- If you sort by "City," all employees in the same city will be grouped perfectly together.